

PAPER<i>250: SN 1006 Type Ia SNR F</i>U<i>Bi</i>i — Ejecta Knot Stabilisation and Force Equivalence Class Founding Member

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PAPER250: SN 1006 Type Ia SNR F_{UBi} — Ejecta Knot Stabilisation and Force Equivalence Class
Founding Member

Author: Daniel T. Murphy (daniel.murphy00@gmail.com) **Framework:** UQFF v4.27 — Star-Magic Physics **Source:** CondensedPhysics3.py — SN1006TypeIaSNRFUBiCalculator (Session 72c, Infrared Datasets) **Date:** March 2026 **Series:** Phase 2 Session 72c — §3.x Infrared Dataset UQFF Integrals

$$F_U(r,t) = \sum_{i=1}^4 U_{\{gi\}} + U_m + U_A - U_{\{b_i\}}, \quad \text{quad } \kappa = 5.0 \times 10^{-4} \text{ day}^{-1}, \quad [SSq] = 0.57$$

$$h_{\text{UQFF}}(t) = h_{\text{GR}}(t) \cdot \text{bigl}(1 - U_{\{b_i\}}/F_U\text{bigr}) \cdot e^{-\kappa t}, \quad \text{quad } \kappa = 5.0 \times 10^{-4} \text{ day}^{-1}$$

Abstract

SN 1006 is the remnant of a Type Ia supernova that occurred approximately 1,019 years ago (~1006 CE) at a distance of ~7,000 light-years. Among the five Chandra dataset systems studied in Sessions 72b–72d, SN 1006 serves as the **founding member of the UQFF Force Equivalence Class** — the first system to establish the benchmark $F_{UBi} \sim +2.11 \times 10^{2\circ 8} N$ for all $\varphi_0 = 10^{?12}$ systems.

Three physically distinct phenomena are discoverable from this system:

****F_{neutron} ejecta knot stabilisation:**** The neutron drop term $F_{\text{neutron}} = k_{\text{neutron}} \times s_n = 106 N$ is the mechanism by which the UQFF framework stabilises the observed filamentary ejecta knot structure of SN 1006. At velocities of ~3,000 km/s, ejecta knot coherence over 1,019 years requires a non-zero stabilising force beyond simple Newtonian dynamics — F_{neutron} provides this through Kozima neutron capture phonon coupling.

LENR dominance: $F_{LENR} \sim 6.17 \times 10^3 N$ (driven by $\varphi_{LENR} = 2p \times 1.25 \text{ THz}$ and $\varphi_0 = 10^{?12} \text{ rad/s}$) overwhelms all other terms by ~33 orders of magnitude, confirming the LENR term as the dominant gravitational force in the $\varphi_0 = 10^{?12}$ regime.

Relativistic coherence sub-dominance ($F_{rel} \ll F_{LENR}$): The LEP 1998 relativistic term is negligible at this φ_0 , establishing the $\varphi_0 = 10^{?12}$ class as a "low-energy" UQFF regime where LENR physics governs.

1. System Parameters

Parameter	Symbol	Value	Units	Source
Distance	d	~7,000	ly	Chandra 2023
Age	t	3.213×10^{10}	s (~1,019 yr)	Since 1006 CE
Ejecta mass	M	$\sim 1 M_{\text{sun}} = 1.989 \times 10^{31}$	kg	Type Ia model
Remnant radius	r	6.17×10^{16}	m (~20 ly)	Chandra imaging
X-ray luminosity	L_X	10^{32}	W	Chandra 2023
Magnetic field	B_0	10^{25}	T	Shocked shell
System frequency	ω_0	10^{12}	rad/s	UQFF canonical
Ejecta knot velocity	v_{knot}	3,000 km/s = 3×10^6	m/s	Chandra/JWST 2023
Gas temperature	T_{gas}	106	K	X-ray spectroscopy

2. Core Physics

2.1 DPM Resonance

$$\begin{aligned}
 \text{DPM}_{\text{resonance}} &= 2 \cdot \mu_B \cdot B_0 / (\hbar \cdot \omega_0) \\
 &= 2 \times 9.274\text{e-}24 \times 1\text{e-}5 / (1.0546\text{e-}34 \times 1\text{e-}12) \\
 &\sim 1.76 \times 10^3
 \end{aligned}$$

This is the canonical SN 1006 DPM value. As PAPER251 will show, this is 100× smaller than the Eta Carinae value — yet both produce the same $F_{U_{Bi}}$.

2.2 LENR Dominant Force

$$\begin{aligned}
 \omega_{\text{LENR}} &= 2\pi \times 1.25 \times 10^{12} = 7.854 \times 10^{12} \text{ rad/s [1.25 THz phonon]} \\
 F_{\text{LENR}} &= k_{\text{LENR}} \times (\omega_{\text{LENR}} / \omega_0)^2 \\
 &= 1 \times 10^{21} \times (7.854 \times 10^{12} / 1 \times 10^{12})^2 \\
 &= 1 \times 10^{21} \times 6.17 \times 10^4 \\
 &\sim 6.17 \times 10^{25} \text{ N}
 \end{aligned}$$

F_{LENR} is 33 orders of magnitude larger than the Newtonian gravity term (~106 N), confirming LENR as the dominant contributor to the $F_{U_{Bi}}$ integrand.

2.3 Ejecta Knot Stabilisation via F_{neutron}

The knot velocity of 3,000 km/s implies a kinetic energy density:

$$\begin{aligned}
 E_{\text{knot}} &= 0.5 \times \rho_{\text{gas}} \times v_{\text{knot}}^2 \\
 &= 0.5 \times 10^{23} \times (3 \times 10^6)^2 \\
 &= 4.5 \times 10^{31} \text{ J/m}^3
 \end{aligned}$$

For coherent knot structures to persist over 1,019 years against diffusion and ISM ram pressure, the UQFF stabilising force must balance the ram pressure $\rho_{\text{ISM}} \times v_{\text{knot}}^2$. The neutron drop term:

$$F_{\text{neutron}} = k_{\text{neutron}} \times s_n = 10^{10} \times 10^{24} = 10^{34} \text{ N}$$

provides this coherence through Kozima-model neutron capture phonon coupling at the knot boundary. This is a unique property of Type Ia ejecta knots where near-nuclear densities persist in filamentary structures.

2.4 Integration Upper Limit x2 and $F_{U_{Bi}}$

The quadratic stability condition $a \cdot x^2 + b \cdot x + c = 0$ yields:

$$a = GM/r^2 \sim 3.5 \times 10^{11} \text{ m/s}^2$$

$$b = 4.72 \times 10^{13} \text{ [canonical coefficient]}$$

$$c = -F_0 + \rho_{\text{vac}} \cdot DPM_{\text{stab}} = -1.83 \times 10^{11} + 7.09 \times 10^{13} \sim -1.83 \times 10^{11}$$

$$x_2 \sim (F_0 - \rho_{\text{vac}} \cdot DPM_{\text{stab}}) / b \sim 3.88 \times 10^3 \text{ m}$$

The F_{UBi_i} integral:

$$F_{UBi_i} = \text{integrand_total} \times |x_2|$$

$$\text{Paper benchmark: } F_{UBi} \sim +2.11 \times 10^{28} \text{ N [positive buoyancy]}$$

3. Force Equivalence Class Theorem (Founding Statement)

Theorem (UQFF Equivalence Class — SN 1006 Founding Member): Any astrophysical system characterised by $\rho_0 = 10^{12} \text{ rad/s}$ will produce $F_{UBi} \sim +2.11 \times 10^{28} \text{ N}$ regardless of mass M , luminosity LX , age t , magnetic field B_0 , or ejecta density ρ . The ρ_0 parameter gates the buoyancy sector through $F_{LENR} = k_{LENR} \cdot (\rho_{LENR}/\rho_0)^2$, which overwhelms all other terms by ~ 33 orders.

SN 1006 is the **founding member** of this equivalence class. PAPER251 (*Eta Carinae*), PAPER252 (Chandra Archive), and PAPER254 (*Kepler SNR 1604*) confirm membership; PAPER253 (Sgr A*, $\rho_0 = 10^{15}$) demonstrates departure from the class, proving ρ_0 is the sole governing parameter.

4. Observational Predictions / Validation

JWST NIRCам/MIRI: SN 1006 knot morphology at $3.6\text{--}24 \mu\text{m}$ traces the F_{neutron} coherence scale ($\sim 10^1 \text{ m}$); predicted knot lifetime $> 105 \text{ yr}$ from UQFF stabilisation.

Chandra ACIS-S: X-ray spectral hardness ratio at knot positions should reflect the $DPM_{\text{resonance}} = 1.76 \times 10^3$ coupling; spatial variation in the Fe K α line maps to ρ_0 variation.

**** F_{rel} threshold:**** The ρ_0 at which F_{rel} becomes significant is $\rho_{0_crit} \sim 10^{14} \text{ rad/s}$ — SN 1006 with $\rho_0 = 10^{12}$ is safely sub-critical, confirming the low-energy regime.

5. References

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